

GEO SHIELD AI

- **Problem Statement ID – 26192**
- **Problem Statement Title- Flash Flood Prediction System for Hilly Regions using Multi-Source Data**
- **Theme**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID-**
- **Solver@SBU**



GEO SHIELD AI

One smart brain watches rain, soil and slopes — and warns every village BEFORE danger comes.

THE PROBLEM WE SOLVE

- Floods & landslides in the hills give almost NO warning time
- Today's alerts cover a whole district — too big for one small valley
- Every monsoon: people die, homes break, roads blocked for days
- NDRF reaches AFTER the disaster — never before it

OUR IDEA — GEOSHIELD AI

- Joins all data together: rain (IMD + satellite), soil sensors, slope maps, past disaster records
- AI checks flood + landslide danger every 15 min for each 1-km village area
- Warns people automatically: siren, SMS/call & app in local language + safe route map

WHAT MAKES IT UNIQUE

- Village-level warnings, not district average — a first of its kind
- Flood + landslide in ONE system — because they strike together
- Works WITHOUT internet: local sirens + LoRa radio mesh
- Uses govt's own alert system (Sachet) — zero extra cost

TECHNOLOGIES — DATA SOURCES

IMD rain station data

NASA satellite rain

GSI landslide atlas

NRSC Bhuvan DEM

SMAP soil moisture

TECHNOLOGIES — BUILD (languages · frameworks · hardware)

Python

AI: LSTM

FastAPI

PostGIS

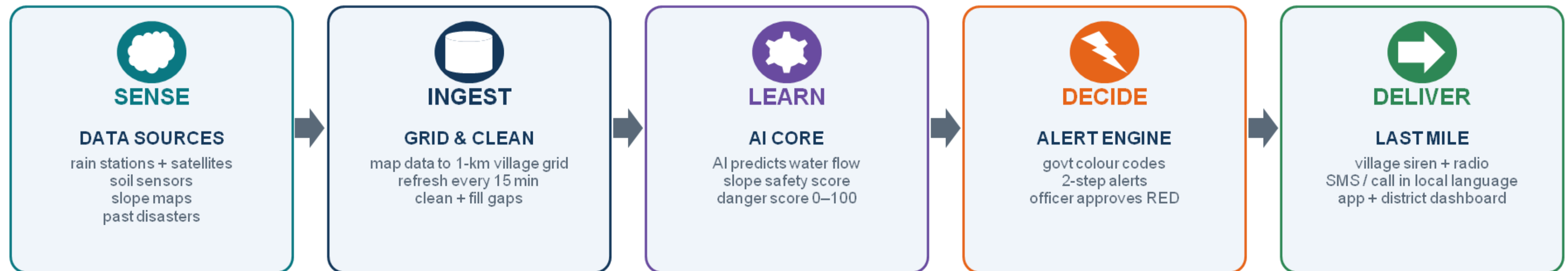
MQTT / LoRaWAN

React dashboard

GSM SMS / IVR

Solar IoT nodes

METHODOLOGY & PROCESS — HOW THE PROTOTYPE WORKS (FLOW)



BUILD & VALIDATION PLAN

1

Teach the AI with 10 years of rain & disaster records

2

Test live in one hill district (e.g., Rudraprayag)

3

Measure: warning hours gained? misses? false alarms?

4

Grow state-by-state with NDRF practice drills

ANALYSIS OF FEASIBILITY & VIABILITY

✓ All data is free & already available

IMD rain, satellite maps (Bhuvan), GSI landslide atlas, NDMA records

✓ Science is proven

Published AI rain models + govt-approved slope methods

✓ Hardware is cheap

one solar sensor box ≈ ₹12–15k; covers a whole valley

✓ Govt systems are ready

NDRF/SDMA control rooms & Sachet alerts we simply plug into

CHALLENGES & RISKS → STRATEGIES TO OVERCOME

Few sensors in far valleys

→

satellite rain + AI fills the gaps

Power & network cuts in storms

→

solar + radio mesh keep working 48 hrs

People ignore false alarms

→

2-step alerts; officer approves final RED

Hard to maintain in hills

→

self-checking boxes + trained village volunteers

IMPACT AND BENEFITS

6–12 hrs

hours of warning — today people get only minutes

1 km²

per village / ward — much finer than district alerts

₹0

extra alert cost — uses govt's existing channels

12 states

fits all notified landslide-prone hill states

IMPACT ON TARGET AUDIENCE: hill villagers & panchayats · NDRF / SDMA · district & state admins

SOCIAL

- First goal: save lives
- People leave calmly on planned routes
- Schools & clinics reopen faster

ECONOMIC

- Less damage to crops, roads & homes
- Risk maps help insurance & safer homes
- Confidence for hill tourism & business

ENVIRONMENTAL

- Guides eco-safe land use in fragile hills
- Less panic digging / blasting after disasters
- Risk maps protect river & forest zones

OPERATIONAL

- NDRF sends teams & boats BEFORE disaster
- One screen for district & state control rooms
- Practice-drill mode keeps people ready

GeoShield AI turns seconds of warning into hours of safety — an affordable digital kavach for India's mountains.

DATASETS & PLATFORMS (all public, all Indian-grade)

- **IMD — real-time rain & forecasts**
AWS/ARG station feeds + QPF products · mausam.imd.gov.in
- **NRSC Bhuvan — terrain & layers**
CartoDEM, slope and watershed boundaries · bhuvan.nrsc.gov.in
- **GSI Bhukosh — landslide inventory**
1:50,000 national landslide susceptibility atlas · bhukosh.gsi.gov.in
- **NASA GPM-IMERG — satellite rain**
30-minute global precipitation estimates · gpm.nasa.gov
- **SMAP / GLDAS — soil moisture**
root-zone moisture for saturation-trigger logic

MODELS & OFFICIAL GUIDELINES

- **Kratzert et al. (2019), HESS**
LSTM for universal, regional & local rainfall–runoff behaviour
- **USGS TRIGRS + Iverson (2000)**
rainfall infiltration & shallow slope-stability science
- **van Westen et al.**
landslide susceptibility mapping — the field standard
- **NDMA guidelines**
early-warning communication & landslide risk management
- **Sachet / CAP protocol**
national common-alerting rails we plug alerts into